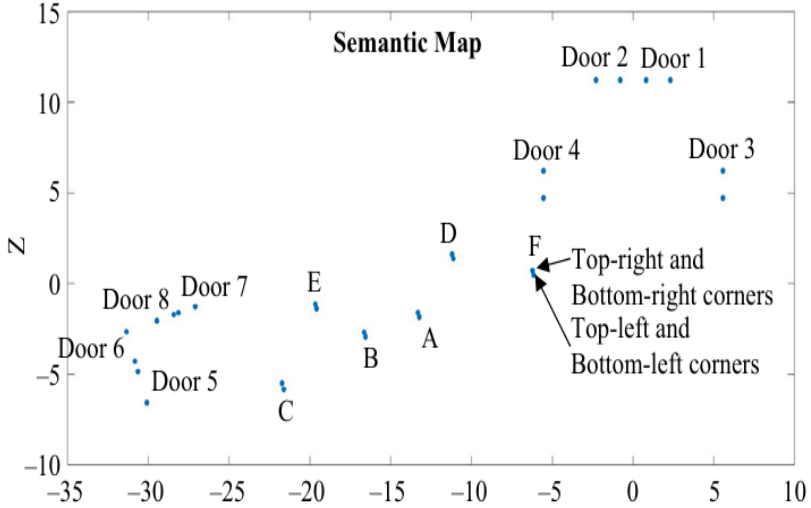
Consider that the map contains several alphabets placed manually, doors, and some other commonly occurring objects, shown in Fig (a). Consider that a live image taken by the robot is as shown in Fig (b). For simplicity, let the characteristic point of the robot used for localization be the same as the camera, which gives $T_R^{cam} = I_{4X4}$ as an identity matrix.



(a)



(b)

TABLE 2.1 Localization using objects detected in a known Semantic Map.

Semantic object	Corner	Observed image coordinates $(u_i^{obs}, v_i^{obs})^T$[a]	Homogenized world coordinates from semantic database (l_i)[b]	Unnormalized image coordinates $(u_i^{proj'}, v_i^{proj'}, w, 1)^T = C T_R^{cam} T_W^R l_i$[c]	Predicted image coordinates $\left(u_i^{proj}, v_i^{proj}\right)^T = \left(\frac{u_i^{proj'}}{w}, \frac{v_i^{proj'}}{w}\right)^T$	Squared error $\left(u_i^{obs} - u_i^{proj}\right)^2 + \left(v_i^{obs} - v_i^{proj}\right)^2$
A	Top-left	(752, 685)	(−13.236, 0.7, −1.8512, 1)			
	Top-right	(802, 685)	(−13.316, 0.7, −1.6039,1)			
	Bottom-right	(802, 752)	(−13.316, 1.05, −1.6039, 1)			
	Bottom-left	(752, 752)	(−13.236, 1.05, −1.8512, 1)			
B	Top-left	(806, 652)	(−16.564, 0.7, −2.9327, 1)			
	Top-right	(842, 652)	(−16.645, 0.7, −2.6855, 1)			
	Bottom-right	(842, 700)	(−16.645, 1.05, −2.6855, 1)			
	Bottom-left	(806, 700)	(−16.564, 1.05, −2.9327, 1)			

TABLE 2.1 Localization using objects detected in a known Semantic Map—cont'd

Semantic object	Corner	Observed image coordinates $(u_i^{obs}, v_i^{obs})^T$	Homogenized world coordinates from semantic database (l_i)	Unnormalized image coordinates $(u_i^{proj'}, v_i^{proj'}, w, 1)^T = CT_R^{cam} T_W^R l_i$	Predicted image coordinates $\left(u_i^{proj}, v_i^{proj}\right)^T = \left(\frac{u_i^{proj'}}{w}, \frac{v_i^{proj'}}{w}\right)^T$	Squared error $\left(u_i^{obs} - u_i^{proj}\right)^2 + \left(v_i^{obs} - v_i^{proj}\right)^2$
D	Top-left	(1302, 762)	(−11.127, 0.7, 1.3679, 1)			
	Top-right	(1378, 762)	(−11.204, 0.7, 1.6057, 1)			
	Bottom-right	(1378, 861)	(−11.204, 1.05, 1.6057, 1)			
	Bottom-left	(1302, 861)	(−11.127, 1.05, 1.3679, 1)			
E	Top-left	(1058, 639)	(−19.591, 0.7, −1.3823, 1)			
	Top-right	(1090, 639)	(−19.669, 0.7, −1.1446, 1)			
	Bottom-right	(1090, 678)	(−19.669, 1.05, −1.1446, 1)			
	Bottom-left	(1058, 678)	(−19.591, 1.05, −1.3823, 1)			
Root mean square error						**23.53**

Calculation of error for a single pose estimate is shown. Optimization is used to get the best pose estimate (localization).
[a]Image: *Y*-axis is positive in the top-down direction, *X*-axis is positive in the left-right direction, and origin at top-left.
[b]World: *Y*-axis is positive facing the ground. The robot moves at the *XZ* plane.

[c]guessed pose of robot $= T_R^W = \begin{bmatrix} -0.3536 & 0 & -0.9354 & -5.5404 \\ 0 & 1 & 0 & -0.1486 \\ 0.9354 & 0 & -0.3536 & 1.7396 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (this guess is optimized as the problem of localization).

$$T_W^R = \left(T_R^W\right)^{-1} = \begin{bmatrix} -0.3536 & 0 & 0.9354 & -3.5863 \\ 0 & 1 & 0 & 0.1486 \\ -0.9354 & 0 & -0.3536 & -4.5673 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$T_R^{\text{cam}} = I_{4\times4} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (camera's coordinate system is used as the robot's coordinate system).

$C = \begin{bmatrix} 1421.28 & 0 & 865.24 & 0 \\ 0 & 1420.64 & 528.13 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (known Calibration matrix, extended to 4 × 4).

Step 1:

Let's choose the homogenized world coordinate for Object D, top-left corner:

$$l_i = \begin{bmatrix} -11.127 \\ 0.7 \\ 1.3679 \\ 1 \end{bmatrix}$$

Apply the robot pose transformation

$$T_W^R = \begin{bmatrix} -0.3536 & 0 & 0.9354 & -3.5863 \\ 0 & 1 & 0 & 0.1486 \\ -0.9354 & 0 & -0.3536 & -4.5673 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

homogenized world coordinate for Object D, top-left corner

$$r_i = l_i \; X \; T_W^R$$

$$r_i = \begin{bmatrix} -11.127 \\ 0.7 \\ 1.3679 \\ 1 \end{bmatrix} X \begin{bmatrix} -0.3536 & 0 & 0.9354 & -3.5863 \\ 0 & 1 & 0 & 0.1486 \\ -0.9354 & 0 & -0.3536 & -4.5673 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$r_i = \begin{bmatrix} 1.6307 \\ 0.8486 \\ 5.3580 \\ 1 \end{bmatrix}$$

Step 2: Apply Camera Calibration Matrix

$$C = \begin{bmatrix} 1421.28 & 0 & 865.24 & 0 \\ 0 & 1420.64 & 528.13 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Multiply C by robot frame point:

$$u_{un\ normal}^{proj} = C \quad X \quad r_i$$

$$u_{un\ normal}^{proj} = \begin{bmatrix} 1421.28 & 0 & 865.24 & 0 \\ 0 & 1420.64 & 528.13 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad X \begin{bmatrix} 1.6307 \\ 0.8486 \\ 5.3580 \\ 1 \end{bmatrix}$$

$$u_{un\ normal}^{proj} = \begin{bmatrix} 6957.938 \\ 4.37.931 \\ 5.3580 \\ 1 \end{bmatrix}$$

Step 3:

This result is the normalized image coordinate.

$$u_i^{proj} = \frac{6957.938}{5.3580} \approx 1299.3$$

$$v_i^{proj} = \frac{4037.931}{5.3580} \approx 754.5$$